

IT-304(GS)

B.Tech. III Semester

Examination, December 2025

Grading System (GS)

Object Oriented Programming and Methodology

Time: Three Hour

Maximum Marks: 70

Note: i) Attempt any five questions.

ii) All questions carry equal marks.

- 1 (a) What are the unique advantages of an object oriented programming paradigm.
- (b) What is Inheritance? What is the importance of Inheritance? Differentiate between multilevel and multiple inheritance with suitable example.
- 2 (a) Explain about call by reference and return by reference with program.
- (b) How is data Encapsulation is different from Information Hiding? Explain with example.
- 3 (a) What is the difference between overloading and overriding? How is binding used in polymorphism? Explain with example.
- (b) Explain function overloading with example.

- 4 (a) How is polymorphism achieved at?
i) Compile Time
ii) Run Time
- (b) Write short note on: Disk file I/O with stream And Data Hiding
- 5 (a) What are exceptions? What are the difference between synchronous and asynchronous exceptions?
- (b) What is friend function? Explain application of friend function with suitable example.
- 6 (a) Write C++ programs to validation of examination marks is to be done. The range of exam marks is between 65 and 100 (both inclusive). This is to be done by including a validate method in student class.
- (b) What is Inline function. What are the advantages of inline function.
- 7 (a) Define operator overloading ? Explain How to overload unary operator and binary operator.
- (b) Explain : i) new and delete keyword ii) This pointer.
- 8 (a) Explain virtual function and write its purpose.

- (b) What do you mean by stream class and stream errors.